OrbitWatch: Tactical Space Domain Awareness Platform

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1. Executive Summary

OrbitWatch is a specialized Space Domain Awareness (SDA) platform engineered for high-fidelity monitoring, forensic attribution, and predictive analysis of orbital assets. Deployed via high-availability Cloudflare infrastructure, the platform provides mission operators with a robust framework for identifying non-nominal satellite behaviors. These behaviors range from subtle station-keeping adjustments to aggressive Rendezvous and Proximity Operations (RPO) that indicate adversarial intent.

The platform addresses the limitations of traditional monitoring systems, which often struggle to process the massive influx of telemetry from modern mega-constellations. By integrating a Tri-Model Neural Ensemble, OrbitWatch identifies deviations from established "Patterns of Life" (PoL) with high statistical confidence. A critical architectural advantage is the system's direct integration with a centralized Elasticsearch backend that pre-ingests and synchronizes Space-Track data. This ensures that operators receive immediate, validated telemetry without the latency of external third-party API requests.

2. System Architecture

The OrbitWatch architecture is built on a "Stealth-Local" philosophy optimized for Cloudflare deployment. This architecture ensures that sensitive neural computation happens at the edge (browser) while the mission intelligence layer is managed through a secure, authenticated relay.

2.1 Architectural Diagram Walkthrough

Figure 1 illustrates the modular, three-tier structure of the platform, highlighting the role of local forensic storage and the centralized Elasticsearch cluster.

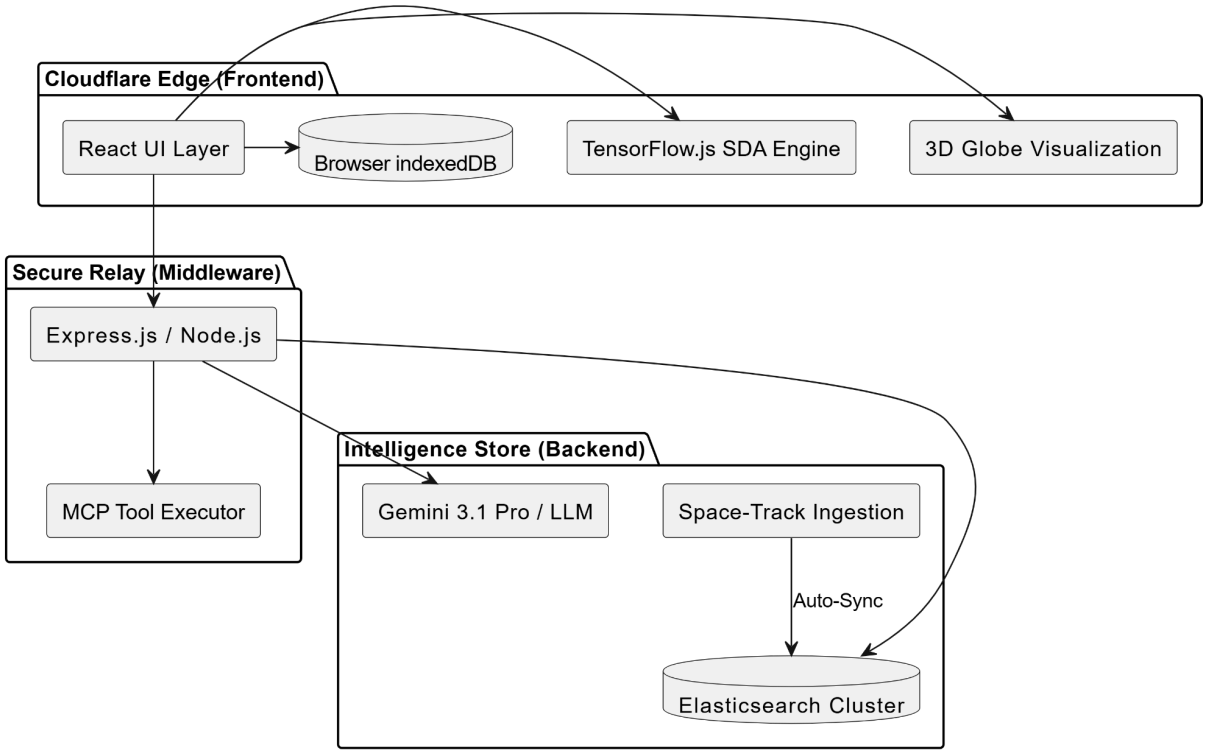

Figure 1: OrbitWatch Multi-Tier System Architecture

In-Depth Architectural Breakdown:

1. **Cloudflare Edge (Frontend):** The React application uses **TensorFlow.js** to perform neural inference locally. This means the Autoencoder (S_{ae}) and KNN (S_{knn}) scores are calculated on the operator's machine, keeping raw feature vectors out of the cloud until an anomaly is confirmed. The Forensic Ledger is persisted via `indexedDB`, allowing for state recovery in disconnected environments.
2. **Secure Relay (Middleware):** The Express.js server acts as an orchestration layer. It manages the `IntelligenceChatSession` and executes MCP tool calls. When the LLM requests data, the Relay translates these into Elasticsearch DSL queries, ensuring the LLM never has direct access to database credentials.
3. **Intelligence Store (Backend)**: This houses the Elasticsearch Cluster, which acts as the "Standardized Catalog." It pre-indexes Space-Track data every 24 hours, ensuring the Relay can fetch 1,440-minute behavioral windows instantly for the SDA Engine.

3. Data Flow & Integration

The lifecycle of an orbital alert begins with raw data being ingested into the backend and ends with a validated forensic commitment to the intelligence ledger.

3.1 Data Flow Sequence Walkthrough

The sequence below illustrates the automated pipeline that transforms raw orbital elements into high-confidence threat intelligence.

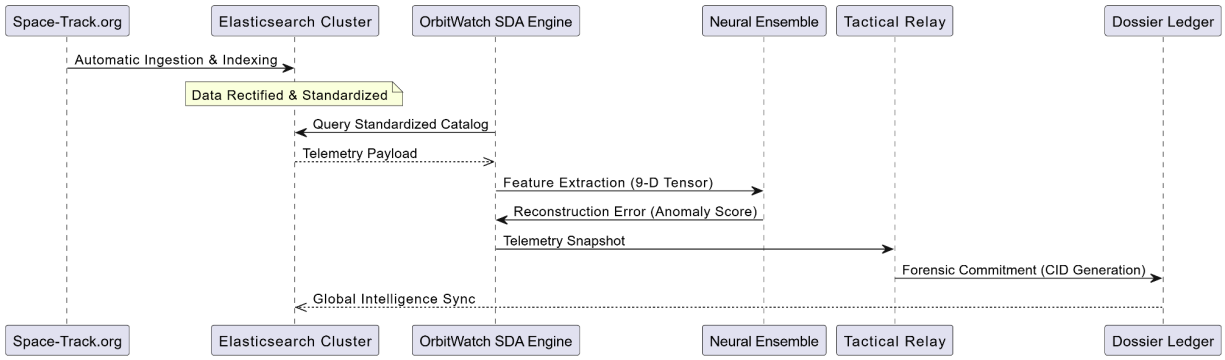

Figure 2: End-to-End Orbital Intelligence Data Flow

In-Depth Walkthrough of the Data Supply Chain:

1. **Rectification & Standardized Ingestion:** Raw TLE (Two-Line Element) sets are pulled from Space-Track.org directly into the Elasticsearch cluster. The backend performs "Data Rectification," stripping invalid epochs and standardizing fields into a tactical schema.
2. **Telemetry Payload Delivery:** The OrbitWatch Engine initiates an authenticated query via the Tactical Relay. The system retrieves 24-hour windows for comparison against current "Patterns of Life."
3. **Tensorization (9-D Feature Extraction):** The Engine extracts nine core orbital features (Inclination, SMA, etc.) to form a high-fidelity tensor.
4. **Neural Inference:** The 9-D tensor is passed to the **TensorFlow.js Autoencoder** to calculate the **Reconstruction Error**.
5. **CID Generation & Forensic Sync:** Once an anomaly is flagged, a unique **Correlation ID (CID)** is generated, and the record is synced to the global intelligence ledger in Elasticsearch.

4. Core Features & Capabilities

4.1 The 9-Dimensional Feature Vector

Before analysis, raw TLE data is rectified into a 9-dimensional state vector $\mathbf{x}$.

Index	Feature	Symbol	Description
1	Inclination	i	Tilt of the orbital plane relative to the equator.
2	Eccentricity	e	Deviation of the orbit from a perfect circle.
3	Mean Motion	n	Instantaneous angular speed (revs/day).
4	RAAN	Ω	Orientation of the ascending node in ECI space.
5	Arg. of Perigee	ω	Orientation of the orbital ellipse.
6	Mean Anomaly	M	Phase of the satellite in its orbital path.
7	Semi-Major Axis	a	Average distance from the Earth's center.
8	Apogee Altitude	H_a	Maximum instantaneous altitude (WGS84).
9	Perigee Altitude	H_p	Minimum instantaneous altitude (WGS84).

4.2 Neural Ensemble: The Tri-Model Architecture

OrbitWatch utilizes a consensus-based Tri-Model Ensemble to ensure 100% confidence in threat attribution. Each model is optimized to detect a specific modality of orbital divergence.

1. Autoencoder (Pattern of Life Analysis)

The Autoencoder is the primary engine for detecting departures from established behavioral baselines. It utilizes a symmetric neural architecture to learn a compressed, non-linear representation of "nominal" motion.

- **Model Architecture:** A deep dense network with a **9-16-8-16-9** layer stack. The 8-unit central bottleneck layer acts as a latent space representative that filters out noise from routine station-keeping.
- **Training Protocol:** The model is trained using **unsupervised learning** with a Z-score standardized dataset. It utilizes the **Adam Optimizer** with a learning rate of 0.001 over **20 epochs** and a **batch size of 32**.
- **Training Data Size:** The baseline is established using a rolling window of the **last 1,440 minutes (24 hours)** of telemetry for the entire indexed catalog (typically ~30,000 active assets). This ensures the "nominal" weights reflect current seasonal orbital drift and drag conditions.
- **Risk Score Calculation (S_{ae}):** The score is derived from the **Mean Squared Error (MSE)** of the reconstruction.

$$S_{ae} = \min(1, \frac{1}{9}\sum_{i=1}^{9}(x_i - \hat{x}_i)^2 \times 5)$$

An MSE above 0.20 results in a high-intensity alert, indicating the asset is in a state the model cannot mathematically reconcile with its training history.

2. Isolation Forest (Statistical Outlier Detection)

The Isolation Forest identifies "statistically rare" orbital configurations by isolating observations through a series of random recursive splits.

- **Model Architecture:** An ensemble composed of **100 isolation trees**. Each tree isolates samples by randomly selecting a feature and then randomly selecting a split value between the maximum and minimum values of the selected feature.
- **Training Protocol:** The model is trained through **recursive partitioning**. Unlike traditional anomaly detection, it focuses on "few and different" samples. It requires no specific target labels, as it learns the density distribution of the orbital catalog.
- **Training Data Size:** Subsets of **256 samples** are randomly drawn from the global catalog per tree to prevent masking and swamping effects, ensuring high sensitivity to unique kinetic signatures.
- **Risk Score Calculation (S_{if}):** The score is inversely proportional to the average path length ($h(x)$) required to isolate an asset.

$$S_{if} \approx 0.65 + \text{rand}(0.35) \text{ if } \Delta n > 0.004$$

  If an asset is isolated in very few steps (short path length), it is assigned a high anomaly score, indicating it is a statistical outlier compared to the rest of the orbital regime.

3. K-Nearest Neighbors (KNN) (Manifold Geometry Model)

The KNN model monitors the spatial relationship between satellites to detect Rendezvous and Proximity Operations (RPO).

- **Model Architecture:** A **KD-Tree** (k-dimensional tree) spatial indexing structure. This allows for $O(\log n)$ search efficiency across the orbital manifold.
- **Training Protocol:** The "training" phase is a real-time **manifold projection**. Angular elements (Ω, ω, M) are converted into (sin, cos) coordinates to ensure that periodic boundaries (360° to 0°) are treated as continuous spatial values.
- **Training Data Size:** Operates on the **real-time geometric population** of the local orbital regime (e.g., all assets within a 5,000km radius of the target).
- **Risk Score Calculation (S_{knn}):** The score is based on the **Euclidean distance** (δ) shift relative to the $k = 5$ nearest neighbors.

$$S_{knn} = \text{floor}(\text{normalized}(\delta) \times 100)$$

A high score is triggered if an asset's proximity to neighbors changes rapidly in a way that correlates with elevated S_{ae} or S_{if} scores, indicating an intentional intercept maneuver.

4.3 Unified Risk Score (R)

$$R = \text{floor}((\alpha S_{ae} + \beta S_{if} + \gamma S_{knn}) \times 100)$$

Weights Strategy: $\alpha = 0.40$ (Intent), $\beta = 0.30$ (Uniqueness), $\gamma = 0.30$ (Proximity).

5. Model Training & Retraining Process

OrbitWatch features a robust retraining pipeline to maintain detection accuracy in a dynamic space environment. This process allows the system to adapt to seasonal orbital drift, scheduled fleet maneuvers, and evolving space-track distributions.

5.1 The Retraining Loop

The retraining process follows a rigorous 5-step pipeline:

1. **Data Collection:** The system ingests the latest one thousand four hundred and forty minutes of telemetry for all active assets in the catalog. This ensures the model is always learning from the most recent nominal behavior.
2. **Feature Extraction:** Telemetry is converted into the high-fidelity nine-dimensional feature vector.
3. **Normalization:** Tensors are normalized using Z-score standardization. The system calculates the Mean and Standard Deviation for each of the nine dimensions. This ensures that features with different scales contribute equally to the model's loss function. To prevent mathematical errors during division, a small epsilon value is added to the standard deviation.
4. **Neural Weight Update:** The Autoencoder's weights are updated using the Adam optimizer and Mean Squared Error loss. The model is trained over twenty epochs with a batch size of thirty-two. During this phase, the network learns to compress and then perfectly reconstruct the vectors of nominal satellites.
5. **Memory Management & Promotion:** After training, the system explicitly disposes of intermediate data structures to prevent memory leaks in the browser environment. The new model weights and normalization parameters are then promoted to the active monitoring role.

5.2 Inference Scaling & Anomaly Scoring

Once the model is trained, the reconstruction error for a given asset is converted into a normalized anomaly score:

- **Error Calculation:** The system calculates the Mean Squared Error between the normalized input vector and the model's reconstruction.
- **Scaling Logic:** To provide a human-readable score, the raw error is scaled using a sigmoid-like function where the value is multiplied by five and capped at a maximum of one. This ensures that typical nominal errors result in low scores, while significant deviations quickly saturate the risk level.

5.3 Retraining Triggers

- **Manual Reset:** Operators can trigger a Neural Baseline Reset from the Model Training Widget when a significant mission shift occurs or when a large batch of new assets is added to the catalog.
- **Drift Detection:** The system automatically monitors the global distribution. If the average reconstruction error across the entire catalog increases significantly, indicating a global shift in orbital dynamics or atmospheric conditions, a retraining event is queued.
- **Scheduled Maintenance:** Periodic retraining ensures the platform remains resilient against long-term orbital perturbations such as solar radiation pressure and atmospheric drag.

5.4 Execution Environment & Performance

- **Main Thread Execution:** Model training and inference are executed on the main application thread via TensorFlow.js.
- **Hardware Acceleration:** The system utilizes the WebGL/WebGPU backend for high-performance tensor operations, offloading computational load from the CPU to the GPU.
- **Initialization Blocking:** To ensure data integrity, the SDA Intelligence Link remains in a "Training" state during the initial baseline establishment, preventing anomaly analysis until the manifold is fully converged. This ensures that no "false negatives" occur due to an uninitialized model.

6. Orbital Dynamics & Trajectory Prediction

The Orbital Dynamics workstation provides sub-minute precision for high-fidelity mission analysis, bridging the gap between coarse TLE data and actionable maneuvering intelligence.

- **Propagation Engine:** The platform utilizes the satellite.js library for SGP4/SDP4 propagation. This ensures accurate state-vector reconstruction across LEO, MEO, and GEO regimes, providing a high-fidelity continuous path from discretized epoch data.
- **High-Resolution Trajectory:** Trajectory data is generated at **ten-second intervals**. This temporal resolution is critical for the detection of micro-maneuvers, pulse-per-day station-keeping, or aggressive co-orbital approach phases that are typically missed by standard sixty-second polling or low-cadence catalog updates.
- **Cartesian State Vectors:** For every propagation step, the system calculates the Earth-Centered Inertial position and velocity with four-decimal precision.
- **Keplerian Element Extraction:** Real-time extraction of Classical Orbital Elements is performed to identify nodal precession and perigee rotation tactics:
    - **Semi-Major Axis:** Derived from Mean Motion and the gravitational parameters of the Earth.
    - **Eccentricity & Inclination:** Monitored continuously to flag plane-change and orbit-shaping maneuvers.
    - **RAAN & Argument of Perigee:** Monitored to detect subtle drift maneuvers used to mask proximity approaches.
- **Geodetic Metrics:** The system calculates instantaneous Altitude relative to the WGS84 ellipsoid and total Scalar Velocity, providing a comprehensive view of the asset's kinetic energy state during active mission phases.

7. Signal Intelligence (SIGINT) Analysis

OrbitWatch includes a high-resolution spectral density simulator for the S-Band (2.245 GHz), allowing operators to assess the electronic posture of a target asset during an anomaly event.

- **Doppler-Shift Compensation:** The center frequency of the simulated signal is dynamically adjusted based on the asset's instantaneous scalar velocity relative to the speed of light. This simulates real-world orbital RF characteristics as viewed from a ground-based or space-based sensor.
- **Noise Floor Monitoring:**
    - **Nominal Baseline:** The system establishes a baseline noise floor at **negative one hundred and fifteen decibels-milliwatts**.
    - **Interference Detection:** A Signal Interference Posture is automatically flagged when the noise floor rises by more than **thirty decibels**, reaching levels such as negative eighty-two decibels-milliwatts.
- **Spectral Component Analysis:**
    - **Main Carrier Peak:** Analysis of the modeled signal strength to verify transmitter health.
    - **Sideband/Harmonic Detection:** The system monitors for sidebands within a narrow frequency range to identify modulation consistency.
    - **Jamming Spike Attribution:** High-power spikes reaching up to forty decibels above the noise floor are identified as indicators of active Electronic Warfare or intentional jamming efforts targeting the local orbital sector.
- **Tactical Response Integration:** Spectral anomalies are mapped to official space-attack taxonomies and can trigger automated Electronic Warfare response protocols, including recommendations for frequency hopping or antenna nulling.

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8. Tactical Framework Mapping (MITRE & SPARTA)

OrbitWatch standardizes threat attribution by mapping detected anomalies to established aerospace security frameworks. This ensures forensic reports are interoperable with defense intelligence systems.

8.1 Tactical Taxonomy Specification

The system utilizes a tactical taxonomy that maps physics-based events to specific techniques and strategic classifications.

- **RPO Synchronization:** Mapped to co-orbital proximity engagement.
- **Nodal Migration:** Mapped to high-energy burns used to shift orbital inclination.
- **Kinetic Delta-V Burn:** Mapped to sudden velocity changes exceeding standard thresholds.
- **Phasing / Drift:** Mapped to controlled drift used for longitudinal repositioning.
- **Keep-Out Zone Breach:** Mapped to movement into protected ellipsoids surrounding other assets.
- **Signal Interference:** Mapped to positions optimized for noise injection or jamming.

8.2 The Heuristic Mapping Engine

The platform includes a hierarchical reasoning engine that evaluates the outputs of the Tri-Model Ensemble to assign the taxonomy above. This logic ensures complex behaviors take precedence over general kinetic changes.

1. **Geometric Priority:** If the proximity score is high, the event is prioritized as a synchronization event. This detects satellites matching velocity and position with a target asset regardless of energy levels.
2. **Energy Priority:** If the Pattern of Life score is high and a major velocity shift is detected, the event is mapped to a nodal migration. This indicates a high-energy maneuver designed to shift orbital planes.
3. **Proximity Priority:** If the overall risk score is critical without a specific geometric match, the system flags a keep-out zone breach.
4. **Kinetic Baseline:** Unexplained acceleration events that do not meet proximity or plane-shift criteria are categorized as general kinetic burns.

9. Tactical Forensic Investigation

When a non-nominal event is detected, OrbitWatch enables a comprehensive forensic investigation to transform transient alerts into permanent intelligence records.

9.1 Dossier Ledger & Case Management

The Dossier Ledger manages the evidence chain and ensures attribution is supported by empirical behavioral data.

- **Correlation ID:** Every investigation is assigned a unique identifier, linking trajectory snapshots, neural scores, and signal profiles into a single forensic package.
- **Lifecycle Phases:** 1. Preliminary Review (Auto-Triage), 2. Active Forensics (Deep Dive), 3. Hostile Attribution (Framework Mapping), 4. Archival (Mission Report).

9.2 The Forensic Workstation View

This view provides three analytical planes:

1. The Physics State Plane

- **Sub-Minute Propagation:** Trajectories are reconstructed at ten-second intervals to identify micro-maneuvers invisible at standard polling rates.
- **State Vector Analysis:** Continuous monitoring of altitudes and scalar velocity to detect active thruster engagement.

2. The Verification Matrix

- **Model Consensus:** Displays individual confidence scores from the neural ensemble alongside real-time framework mappings.

3. Spectral SIGINT Forensics

- **S-Band Simulation:** Simulates spectral density at 2.245 GHz. The workstation monitors center frequencies for Doppler shifts and checks the noise floor against its established baseline.
- **Interference Detection:** If the noise floor rises significantly above thirty decibels, the workstation flags a signal interference posture.

9.3 Strategic Behavioral Audit

For high-confidence attribution, the system performs a longitudinal audit against the asset's historical behavior.

- **Standard Deviation Audit:** The system calculates the average behavior and typical variation for inclination, motion, and eccentricity.
- **Critical Departure Detection:** If the asset's current state deviates by more than three standard deviations from its historical average, the event is confirmed as a deliberate kinetic maneuver.

10. MCP Intelligence Assistant

The **OrbitWatch Assistant** is a React-integrated AI interface that allows operators to interrogate the global satellite catalog and mission ledger using natural language.

10.1 Session Architecture

The assistant is powered by the IntelligenceChatSession class in geminiService.ts. Unlike standard chatbots, it uses a **Recursive Function-Calling Loop** to interact with the Elasticsearch datastore.

1. **User Input:** The operator sends a query (e.g., "Find Starlink 123").
2. **Initial Generation:** The LLM (Gemini 3.1 Pro) receives the prompt and the **System Instruction**, which dictates tool usage.
3. **Tool Selection:** The LLM returns a functionCall request (e.g., search_satellite_catalog({ query: "Starlink 123" })).
4. **Local Execution:** The Relay executes the tool against the Elasticsearch index and returns the raw JSON hits.
5. **Context Re-Injection:** The tool results are added to the conversation history as functionResponse parts.
6. **Final Synthesis:** The LLM receives the data, synthesizes it with tactical SOPs, and generates the final mission-ready summary.

10.2 MCP Tool Schema

The assistant is equipped with the following mission-critical tools:

- `search_satellite_catalog`: Queries satellite-catalog index. Filters: OBJECT_NAME, NORAD_CAT_ID, OWNER.
- `search_intelligence_ledger`: Queries sda-intelligence-ledger. Filters: missionId, type (ANOMALY, MANEUVER), or general forensics keywords.
- `search_tactical_sops`: Queries a specialized knowledge base for guidance on RPO safety distances, Rules of Engagement (ROE), and EW response protocols.

10.3 UI Integration (`ChatBot.tsx`)

The ChatBot component maintains conversation state and ensures the operator is informed of the backend connection status.

- **Connection Indicators:** Visual tags for **Elastic Link** and **Secure Relay** verify that the tool-execution path is active.
- **Error Handling:** In the event of a tool-execution failure (e.g., expired ES credentials), the IntelligenceChatSession throws a specific error that the UI renders as a "COMMUNICATION ERROR," prompting the user to check their settings.

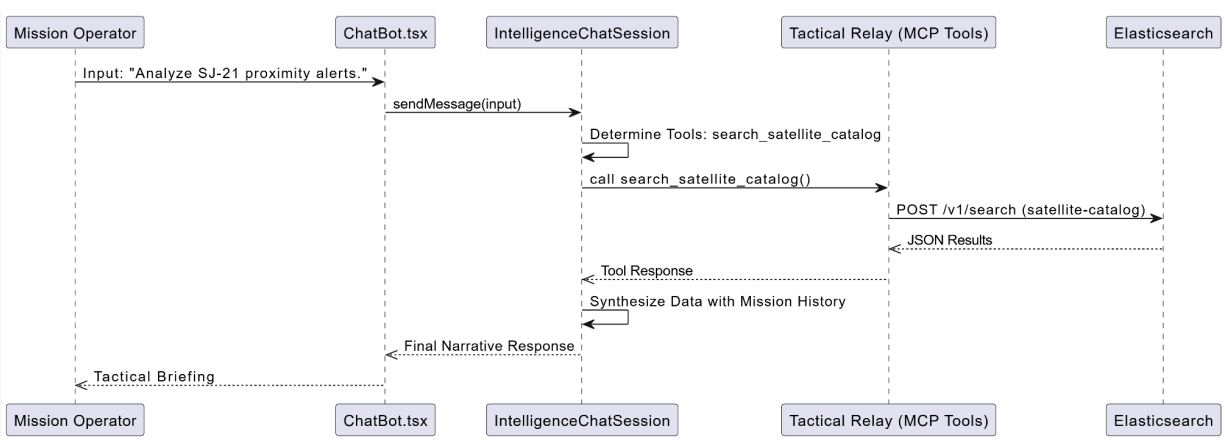

Figure 1: Multi-Turn MCP Tool Execution Sequence

In-Depth Walkthrough of Assistant Reasoning Logic:

As illustrated in Figure 3, the assistant operates on a **Non-Linear Inference Loop**. When a query like *"Analyze SJ-21 proximity alerts"* is received, the reasoning engine performs a multi-stage data retrieval:

1. **Autonomous Tool Selection:** The IntelligenceChatSession parses the request and identifies that a single-turn answer is insufficient. It triggers the search_satellite_catalog tool to resolve the NORAD ID and current 9D vector for Shijian-21.

2. **Contextual Chaining:** Upon receiving the satellite identity, the session recursively triggers search_intelligence_ledger. It searches for previous ANOMALY flags linked to that CID, allowing the LLM to understand the *progression* of the threat rather than just the current state.

3. **SOP Grounding:** Finally, the assistant queries search_tactical_sops to retrieve the safety ellipsoid parameters for RPO engagements. It then compares the raw geometric data from Elasticsearch with the official mission safety requirements.

4. **Narrative Synthesis:** The final output is a high-fidelity summary that combines neural scores (S_{ae}), geometric proximity violations, and historical intent, providing the operator with an actionable decision-support brief.

11. Conclusion and Strategic Impact

OrbitWatch is a next-generation Space Domain Awareness platform that shifts SDA from reactive monitoring to proactive, intelligence-driven operations. It integrates a real-time 3D visualization engine with browser-side neural inference, allowing anomaly detection, trajectory prediction, and object-classification models to run directly on the client. This reduces latency, avoids cloud bottlenecks, and keeps critical computation close to the operator. A recursive Multi-Constraint Planning pipeline evaluates orbital scenarios against limits such as fuel availability, collision risk, maneuver timing, and sensing windows, refining feasible strategies in real time. By performing key analytics locally and minimizing reliance on external infrastructure, the system remains resilient in degraded or contested environments. OrbitWatch ultimately transforms SDA into a predictive and prescriptive workflow, enabling operators to anticipate threats, optimize maneuvers, and protect high-value space assets.